

Glossary of NHC Terms

Advisory:

Official information issued by tropical cyclone warning centers describing all tropical cyclone watches and warnings in effect along with details concerning tropical cyclone locations, intensity and movement, and precautions that should be taken. Advisories are also issued to describe: (a) tropical cyclones prior to issuance of watches and warnings and (b) subtropical cyclones.

Best Track:

A subjectively smoothed path, versus a precise and very erratic fix-to-fix path, used to represent tropical cyclone movement. It is based on an assessment of all available data.

Center:

The vertical axis or core of a tropical cyclone. It is usually determined by cloud vorticity patterns, wind, and/or pressure distributions.

Center/Vortex Fix:

The location of the center of a tropical or subtropical cyclone obtained by reconnaissance aircraft penetration, satellite, radar, or synoptic data.

Central North Pacific Basin:

The region north of the Equator between 140W and the International Dateline. The Central Pacific Hurricane Center (CPHC) in Honolulu, HI is responsible for tracking tropical cyclones in this region.

Cyclone:

An atmospheric closed circulation rotating counter-clockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.

Eastern North Pacific Basin:

The region north of the Equator east of 140W. The National Hurricane Center in Miami, FL is responsible for tracking tropical cyclones in this region.

Eye:

The relatively calm center of the tropical cyclone that is more than one half surrounded by wall cloud.

Eye Wall/Wall Cloud:

An organized band of cumiform clouds immediately surrounding the center of a tropical cyclone. Eye wall and wall cloud are used synonymously.

Explosive Deepening:

A decrease in the minimum sea-level pressure of a tropical cyclone of 2.5 mb/hr for at least 12 hours or 5 mb/hr for at least six hours.

Extratropical:

A term used in advisories and tropical summaries to indicate that a cyclone has lost its "tropical" characteristics. The term implies both poleward displacement of the cyclone and the conversion of the cyclone's primary energy source from the release of latent heat of condensation to baroclinic (the temperature contrast between warm and cold air masses) processes. It is important to note that cyclones can become extratropical and still retain winds of hurricane or tropical storm force.

Fujiwhara Effect:

A binary interaction where tropical cyclones within a certain distance (300-750 nm depending on the sizes of the cyclones) of each other begin to rotate about a common midpoint.

Gale Warning:

A warning of 1-minute sustained surface winds in the range 34 kt (39 mph or 63 km/hr) to 47 kt (54 mph or 87 km/hr) inclusive, either predicted or occurring not directly associated with tropical cyclones.

High Wind Warning:

A high wind warning is defined as 1-minute average surface winds of 35 kt (40 mph or 64 km/hr) or greater lasting for 1 hour or longer, or winds gusting to 50 kt (58 mph or 93 km/hr) or greater regardless of duration that are either expected or observed over land.

Hurricane/Typhoon:

A warm-core tropical cyclone in which the maximum sustained surface wind (using the U.S. 1-minute average) is 64 kt (74 mph or 119 km/hr) or more. The term hurricane is used for Northern Hemisphere cyclones east of the International Dateline to the Greenwich Meridian. The term typhoon is used for Pacific cyclones north of the Equator west of the International Dateline.

Hurricane Local Statement:

A public release prepared by local National Weather Service offices in or near a threatened area giving specific details for its county/parish warning area on (1) weather conditions, (2) evacuation decisions made by local officials, and (3) other precautions necessary to protect life and property.

Hurricane Season:

The portion of the year having a relatively high incidence of hurricanes. The hurricane season in the Atlantic, Caribbean, and Gulf of Mexico runs from June 1 to November 30. The hurricane season in the Eastern Pacific basin runs from May 15 to November 30. The hurricane season in the Central Pacific basin runs from June 1 to November 30.

Hurricane Warning:

An announcement that hurricane conditions (sustained winds of 74 mph/64 kt or higher) are expected somewhere within the specified coastal area. Because hurricane preparedness activities become difficult once winds reach tropical storm force, the hurricane watch is issued 36 hours in advance of the anticipated onset of tropical-storm-force winds.

NOTE: Lead time was changed for 2011 season.

Hurricane Watch:

An announcement that hurricane conditions (sustained winds of 74 mph/64 kt or higher) are possible somewhere within the specified coastal area. Because hurricane preparedness activities become difficult once winds reach tropical storm force, the hurricane watch is issued 48 hours in advance of the anticipated onset of tropical-storm-force winds.

NOTE: Lead time was changed for 2011 season.

Post-storm Report:

A report issued by a local National Weather Service office summarizing the impact of a tropical cyclone on its forecast area. These reports include information on observed winds, pressures, storm surges, rainfall, tornadoes, damage and casualties.

Preliminary Report:

A report summarizing the life history and effects of an Atlantic or eastern Pacific tropical cyclone. It contains a summary of the cyclone life cycle and pertinent meteorological data, including the post-analysis best track (six-hourly positions and intensities) and other meteorological statistics. It also contains a description of damage and casualties the system produced, as well as information on forecasts and warnings associated with the cyclone. NHC writes a preliminary report on every tropical cyclone in its area of responsibility.

Present Movement:

The best estimate of the movement of the center of a tropical cyclone at a given time and given position. This estimate does not reflect the short-period, small scale oscillations of the cyclone center.

Probability of Tropical Cyclone Conditions:

The probability, in percent, that the cyclone center will pass within 50 miles to the right or 75 miles to the left of the listed location within the indicated time period when looking at the coast in the direction of the cyclone's movement.

Rapid Deepening:

A decrease in the minimum sea-level pressure of a tropical cyclone of 1.75 mb/hr or 42 mb for 24 hours.

Relocated:

A term used in an advisory to indicate that a vector drawn from the preceding advisory position to the latest known position is not necessarily a reasonable representation of the cyclone's movement.

Saffir-Simpson Hurricane Scale:

Defines hurricane strength by categories. A Category 1 storm is the weakest hurricane (winds 74-95 mph or 64-82 kt); a Category 5 hurricane is the strongest (winds greater than 155 mph or 135 kt). **NOTE:** These are relative terms, because

lower category storms can sometimes inflict greater damage than higher category storms, depending on where they strike and the particular hazards they bring.

Storm Surge:

An abnormal rise in sea level accompanying a hurricane or other intense storm, and whose height is the difference between the observed level of the sea surface and the level that would have occurred in the absence of the cyclone. Storm surge is usually estimated by subtracting the normal or astronomic high tide from the observed storm tide.

Storm Tide:

The actual level of sea water resulting from the astronomic tide combined with the storm surge.

Storm Warning:

A warning of 1-minute sustained surface winds of 48 kt (55 mph or 88 km/hr) or greater, either predicted or occurring, not directly associated with tropical cyclones.

Subtropical Cyclone:

A low pressure system that develops over subtropical waters that initially has a non-tropical circulation but in which some elements of tropical cyclone cloud structure are present. Subtropical cyclones can evolve into tropical cyclones. Subtropical cyclones are generally of two types:

(1) An upper level cold low with circulation extending to the surface and maximum sustained winds generally occurring at a radius of about 100 miles or more from the pressure center.

(2) A mesoscale cyclone originating in or near a frontolyzing zone of horizontal wind shear, with radius of maximum sustained winds generally less than 30 miles. The entire circulation sometimes encompasses an area initially no more than 100 miles in diameter. These generally short-lived, marine cyclones may vary in structure from cold to warm core.

Subtropical Depression:

A subtropical cyclone in which the maximum sustained surface wind speed (using the U.S. 1-minute average) is 33 kt (38 mph or 62 km/hr) or less.

Subtropical Storm:

A subtropical cyclone in which the maximum sustained surface wind speed (using the U.S. 1-minute average) is 34 kt (39 mph or 63 km/hr) or more.

Synoptic Track:

Weather reconnaissance mission flown to provide vital meteorological information in data sparse ocean areas as a supplement to existing surface, radar, and satellite data. Synoptic flights better define the upper atmosphere and aid in the prediction of tropical cyclone development and movement.

Tropical Cyclone:

A warm-core, nonfrontal low pressure system of synoptic scale that develops over tropical or subtropical waters and has a definite organized surface circulation.

Tropical Cyclone Plan of the Day:

A coordinated mission plan that tasks operational weather reconnaissance requirements during the next 1100 to 1100 UTC day or as required, describes reconnaissance flights committed to satisfy both operational and research requirements, and identifies possible reconnaissance requirements for the succeeding 24-hour period.

Tropical Depression:

A tropical cyclone in which the maximum sustained surface wind speed (using the U.S. 1-minute average) is 33 kt (38 mph or 62 km/hr) or less.

Tropical Disturbance:

A discrete tropical weather system of apparently organized convection--generally 100 to 300 nmi in diameter---originating in the tropics or subtropics, having a nonfrontal migratory character, and maintaining its identity for 24 hours or more. It may or may not be associated with a detectable perturbation of the wind field.

Tropical Storm:

A tropical cyclone in which the maximum sustained surface wind speed (using the U.S. 1-minute average) ranges from 34 kt (39 mph or 63 km/hr) to 63 kt (73 mph or 118 km/hr).

Tropical Storm Warning:

An announcement that tropical storm conditions (sustained winds of 39 to 73 mph or 34 to 63 kt) are expected within

the specified coastal area within 36 hours. NOTE:
Lead time was changed for 2011 season.

Tropical Storm Watch:

An announcement that tropical storm conditions (sustained winds of 39 to 73 mph or 34 to 63 kt) are possible within the specified coastal area within 48 hours. NOTE:
Lead time was changed for 2011 season.

Tropical Wave:

A trough or cyclonic curvature maximum in the trade-wind easterlies. The wave may reach maximum amplitude in the lower middle troposphere.

TUTT:

A "TUTT" is a Tropical Upper Tropospheric Trough. A TUTT low is a TUTT that has completely cut-off. TUTT lows are more commonly known in the Western Hemisphere as an "upper cold low". TUTTs are different than mid-latitude troughs in that they are maintained by subsidence warming near the tropopause which balances radiational cooling. TUTTs are important for tropical cyclone forecasting as they can force large amounts of vertical wind shear over tropical disturbances and tropical cyclones which may inhibit their strengthening. There are also suggestions that TUTTs can assist tropical cyclone genesis and intensification by providing additional forced ascent near the storm center and/or by allowing for an efficient outflow channel in the upper troposphere.